University of Limpopo

Faculty of Health Sciences

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Development and Validation of an AI Model for Detection of Pre-cancerous Cervical Lesions at Colposcopy

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1. Glossary of Abbreviations & Terms

Abbreviation/Term	Description (using lay language)
AI	Artificial Intelligence
LLETZ	Large Loop Excision of the Transformation Zone
HPV	Human Papillomavirus
WHO	World Health Organisation
DNA	Deoxyribonucleic Acid
VIA	Visual Inspection with Acetic Acid
PPV	Positive Predictive Value
CIN	Cervical Intraepithelial Neoplasia
LSIL	Low Grade Squamous Intraepithelial Lesion
HSIL	High Grade Squamous Intraepithelial Lesion
SS-AI	Smart Scope AI
AUC	Area Under the Receiver Operator Curve
ASCUS	Atypical Squamous Cells of Undetermined Significance
ASC-H	Atypical Squamous Cells Cannot Exclude High Grade Squamous Intraepithelial Lesion

2. Study Sites

Study Location/s

Site	Address	Contact Person	Phone	Email
Mankweng Provincial Hospital	Mankweng Complex, Houtbosdorp Street, Polokwane, 0727	Dr Baloyi		
Quality Care Private Hospital	83 Grobler Street, Louis Trichardt, 0920	Dr Tigere		

This study will be conducted at both Mankweng Provincial Hospital and Quality Care Private Hospital to ensure inclusion of patients from both public and private healthcare settings, improving data representativeness.

3. Funding and Resources

3.1 Source/s of Funding

This study will be funded by SAfAIDS (Southern Africa HIV & AIDS Information Dissemination Service) and ASCCP (Australian Society for Colposcopy and Cervical Pathology).

4. Introduction and Background

4.1 Lay Summary

The use of Artificial Intelligence (AI) in healthcare is growing rapidly, offering new opportunities to improve patient care and ease the burden on healthcare providers. One area where AI shows significant promise is in the interpretation of colposcopies—a diagnostic procedure used to examine the cervix, vagina, and vulva for signs of disease, particularly cancerous or precancerous changes.

This study aims to develop an AI model to assist in interpreting cervical images taken during colposcopy. We will use images from our local patient population to fine-tune a model that was initially trained on a large, publicly available dataset. By incorporating data from a local population with similar demographics, comorbidities, and clinical characteristics as the intended end users, we hope to improve the model's relevance and accuracy in our setting.

We will conduct a prospective study involving women undergoing Large Loop Excision of the Transformation Zone (LLETZ), cone biopsy procedures and routine papsmears. During these procedures, cervical images will be captured and paired with the corresponding histology and cytology results. These annotated images will then be used to train and validate our AI model.

The performance of the model will be evaluated using both public and locally sourced datasets to determine whether the inclusion of population-specific data leads to meaningful improvements. We will also compare the model's diagnostic accuracy to that of practicing clinicians and collect feedback on the feasibility of implementing this AI-powered colposcopy tool in clinical practice.

Ultimately, this study aims to support the development of a low-cost, AI-enabled colposcopy tool designed for use in low-resource settings. It will also address important concerns such as algorithmic bias, transparency, and equitable access to cervical cancer screening technologies.

4.2 Introduction

Cervical cancer remains a major public health concern in South Africa, where it ranks as the leading cause of cancer related deaths among women [1]. The country reports an age-standardized incidence rate of 35.3 per 100 000 women, nearly three times the global average [2]. There has been a decline in the incidence and mortality trends in many other parts of the world [3], which has been largely attributed to the widespread uptake of the human papillomavirus (HPV) vaccine [4] and the implementation of high-quality screening programs that effectively prevent cervical cancer among unvaccinated, older women [2,3].

In August 2020, the World Health Assembly adopted the Global Strategy to Accelerate the Elimination of Cervical Cancer as a public health problem, which calls for all countries to reduce and sustain cervical cancer incidence to fewer than 4 cases per 100,000 women. This strategy is built on three key pillars, one of them being that 70% of women should be screened with a high-performance test by age 35 [5].

According to the World Health Organisation (WHO) guidelines, primary HPV DNA testing is now recommended as the preferred screening method in all settings, with screening intervals of 5 to 10 years for HPV-negative individuals [6]. A screen-triage-treat approach with acceptable triage method including colposcopy, visual inspection with acetic acid (VIA) or cytology is advised for those who test positive for high-risk HPV. The selection of a triage method should be based on each country's feasibility, training capacity, quality assurance infrastructure, and available resources [7].

Cytology, while historically a cornerstone of cervical screening, presents significant limitations in the South African context due to its multistep process, lack of laboratory capacity, and high turnaround times [8]. VIA, in contrast, offers several advantages: its low-cost, requires minimal infrastructure, and can be offered at the primary healthcare level without laboratory support. However, its major drawback is its subjectivity, with a high false positive rate resulting in overcalling and overtreating [9].

Traditional colposcopy can be a useful triage tool [10]. With adequate training and supervision, colposcopy can be task-shared with nurses to expand access to diagnostic services. South Africa's Department of Health policy recommends that colposcopy be available at primary and district levels, with screening led by nurses and supported by medical officers and specialists as needed [9].

While colposcopy remains a critical tool for diagnosing cervical pre-cancer and cancer, its effectiveness is limited by factors such as inter-observer variability. In one

study, the accuracy of colposcopy impression was reported as 84.5%, with a positive predictive value (PPV) of 66.4% for cervical intraepithelial neoplasia 2 or worse (CIN2+) lesions [11]. Variability in interpreting cervical appearance remains a key challenge, even among experienced clinicians.

Adding to the subjective nature of colposcopy impression, provider training and practice of colposcopy in Africa remains suboptimal. A survey of clinicians from 23 African countries reported that only 12.6% had received prior training in colposcopy, and just 11.7% were performing colposcopy in their current practice [12].

Other limitations to colposcopy in South Africa include a limited number of colposcopy services, which are largely centralised in tertiary-level facilities, leading to long waiting times and few opportunities for non-specialists to develop the necessary skills [13]. Decentralising these services could improve access and result in faster diagnosis of precancerous cervical lesions, ultimately reducing the incidence of cervical cancer. This was demonstrated in a study conducted in South Africa [14]. If implemented properly and at scale, such decentralisation could strengthen cervical cancer prevention efforts and significantly reduce the public health burden and mortality associated with the disease [14].

With the advent of artificial intelligence (AI) in healthcare, one area where it holds significant promise is in the interpretation of colposcopies. AI tools have the potential to detect patterns that may surpass human capabilities in disease diagnosis [15]. Several studies have already demonstrated high accuracy in detecting cervical precancerous lesions using AI-powered image analysis tools, with some models outperforming clinicians' colposcopy impressions [16,17]. Other known benefits of AI include a reduction in cost, time savings and improvement in clinical workflow [10,15].

Although AI has the potential to revolutionize clinical practice, its full impact can only be realized if key challenges are addressed, such as the lack of high-quality medical data, algorithmic bias, data privacy and security concerns, and issues related to transparency with model development [15].

The objective of this study is to develop and validate an AI model using cervical images collected from patients undergoing LLETZ (Large Loop Excision of the Transformation Zone) or cone biopsy procedures. By training the model on images linked to histological outcomes, we aim to improve the accuracy and consistency of colposcopy interpretation. The long-term goal is to support the development of a reliable, low-cost diagnostic tool that can enhance early detection of cervical precancer and cancer, particularly in low-resource settings where specialist expertise and advanced diagnostic infrastructure may be limited.

4.3 Background and Rationale

Artificial intelligence (AI), particularly computer vision techniques using convolutional neural networks (CNNs), has emerged as a promising solution to enhance diagnostic accuracy and consistency in colposcopy. Several studies have demonstrated the potential of AI to detect cervical precancerous lesions with high sensitivity and specificity, sometimes outperforming human colposcopists in certain tasks [18,19]. Despite these advances, a critical limitation of existing models is the reliance on training data that may not be representative of the target population, as well as lack of validation against gold-standard histopathology.

This literature review will critically examine recent advancements in AI-assisted colposcopy, highlight the methodological and ethical limitations of existing work, and identify gaps that our study aims to address—particularly the need for locally relevant, histology-anchored AI models designed for implementation in resource-constrained environments.

Xue et al, in their review on “The Challenges to Development and Application of AI-Guided Colposcopy,” highlighted several barriers. First, they noted that the use of varying types of colposcopy equipment leads to poorly managed datasets and inadequate quality control, recommending the standardization of equipment to address this. Second, they observed that most existing studies are retrospective, relying on static colposcope images rather than dynamic assessments of the cervix, which may compromise diagnostic accuracy. To mitigate this, they advocate for the use of high-quality images and pathology-confirmed diagnoses to improve the reliability and validity of AI models [10].

Given the subjective nature of colposcope impression and its high inter and intra-observer variability, we propose that histological diagnosis from LLETZ or cone biopsy specimens be used as ground truth for the development of the AI algorithm. Relying solely on colposcopy interpretation risks introducing noise into the training data, ultimately reducing the performance of the AI model.

In a retrospective study involving 70 colposcopy videos, 22 693 image frames were extracted and used to train a convolutional neural network (CNN) model. In this study, colposcopy images were correlated with corresponding biopsy results. The model demonstrated impressive performance metrics, with reported accuracy, sensitivity and specificity of 99%, 99.7% and 98.6% respectively. However, several limitations were identified. These included the model’s inability to differentiate concurrent low-grade and high-grade squamous intraepithelial (LSIL and HSIL) within the same cervical lesion; the exclusive use of a single colposcope brand, raising concerns about the model’s performance on images captured with different devices; and limited generalizability due to training on a specific population, which may increase the risk of overfitting and reduce performance when applied to diverse cohorts [20].

A study conducted in China evaluated the diagnostic accuracy of an AI-powered digital colposcopy device, Smart Scope AI (SS-AI), in comparison to colposcopy impression by an experienced clinician. When SS-AI was compared directly to colposcopy, it demonstrated a sensitivity of 52.17% and a specificity of 93.46%. However, when both SS-AI and colposcopy impression were assessed against histopathological findings, SS-AI achieved a sensitivity and specificity of 83.3%, while colposcopy showed the same sensitivity (83.33%) but a notably lower specificity of 50%. This low specificity for colposcopy highlights its tendency to overcall lesions. The relatively lower sensitivity of SS-AI when compared directly to colposcopy may reflect this overestimation bias inherent in clinician assessment. Notably, in this study, the SS-AI screening was performed by a minimally trained nurse, underscoring the devices potential for deployment in low-resource settings with limited specialist availability. Despite SS-AI's promising performance, the study's conclusions are constrained by a small sample size-only 30 histopathologically-confirmed cases. Our study aims to address these limitations by using histological diagnosis from LLETZ or cone biopsy specimens as the gold standard for identifying cervical precancerous lesions using a larger dataset [21].

Algorithmic bias presents a critical challenge in deploying AI systems for clinical diagnostics, particularly in diverse and resource-limited settings. A study published in The Lancet Digital Health evaluated 12 commercially available AI models for tuberculosis (TB) detection on chest X-ray within a South African population. While tools such as Lunit INSIGHTS CXR and Nexus CXR achieved high overall accuracy (AUC >89%), their performance significantly declined in key subgroups, namely older patients, individuals with previous TB, and people living with HIV. This highlights the importance of validating AI models across the full spectrum of real-world patient demographics, particularly where comorbidities are common [22].

These lessons are highly relevant for AI-based colposcopy, where existing models may underperform if not specifically trained and validated on South African cervical image datasets. Our study addresses this gap by building a model grounded in histologically validated cervical images from South African women, ensuring representation and reducing the risk of performance disparities across key subgroups.

5. Study hypothesis/research question/aim(s)

The aim of this study is to develop and validate an AI model for the detection of pre-cancerous cervical lesions at colposcopy.

6. Study Objectives and Outcomes

To assess the diagnostic performance of an AI model for colposcopy trained on colposcopy images to detect High Grade Squamous Intraepithelial Lesion (HSIL) or worse using histopathology as the reference standard.

6.1 Primary Objectives

To assess the diagnostic performance of an AI model for colposcopy trained on colposcopy images in classifying images into normal, LSIL, HSIL or worse, using histopathology as the reference standard.

6.2 Secondary Objectives

i) To compare the diagnostic performance of an AI model trained on a public cervical image dataset versus a model fine-tuned on locally acquired data.
ii) To compare the model's diagnostic accuracy with that of trained colposcopists interpreting the same set of images.
iii) To evaluate the usability and acceptability of a low-cost, smartphone-based AI colposcopy system among healthcare providers.

6.3 Outcomes/Outcome Measures

Objective	Outcome measure	Description and definition	Assessment time point	Justification
Primary	Diagnostic accuracy	Proportion of correct classification of cervical images into Normal, LSIL, HSIL+ by AI model, compared with	At the time histopathology results are available	Histology is the gold standard

		histopathology results		
Secondary	Model performance comparison	Sensitivity, specificity, AUC of AI trained on public vs local datasets	At model testing stage	Evaluates generalizability
Secondary	AI versus clinician accuracy	Percentage agreement, kappa statistics, sensitivity, specificity	Concurrent review	Benchmarking AI against human assessment
Secondary	Usability and acceptability	Provider responses to structured usability survey	After clinical use	Assess feasibility for implementation

7. Study Design

7.1 Study Type & Design & Schedule

This study is designed as a prospective observational cohort study. Eligible participants will be recruited prior to undergoing clinically indicated colposcopy guided LLETZ or cone biopsy procedures as well as routine papsmears. At the time of the procedure, cervical images will be captured using a smart-phone based imaging device. Approximately 500 patients will be recruited, with at least three to four images taken from each patient, yielding approximately 2000 data points. Images will be taken prior to application of acetic acid, after acetic acid, and post application of Lugol's iodine.

These images will be linked to histopathological and cytological outcomes from the corresponding excisional biopsy specimens and cervical smears. Data collection will occur prospectively, with participants followed from the point of image capture through to the availability of histology results. This design allows for the development and validation of an Artificial Intelligence (AI) model capable of predicting cervical pathology based on visual inspection, while ensuring that all

data are obtained in real time and in alignment with standard clinical care pathways.

Data will be collected from two main sites: Mankweng Provincial Hospital (a public tertiary teaching hospital affiliated with the University of Limpopo) and Quality Care Private Hospital (a private healthcare facility in Louis Trichardt, Limpopo). The inclusion of a private hospital site allows the study to capture a broader range of patient demographics and cervical pathology presentations, thereby enhancing generalisability of the AI model.

Data consists of cervical images, histopathology results, cytology results, demographics, clinical impressions and usability feedback. All data will be de-identified.

The total study duration is approximately 6-12 months.

7.2 Standard Care and Additional to Standard Care Procedures

Standard care

Procedure	Time/visit	Amount
Papsmear	Day of procedure	N/A
LLETZ/Cone biopsy	Day of procedure	N/A
Histopathology reporting	6-8 weeks	N/A

Additional to standard care

Procedure	Time/visit	Amount
Application of acetic acid/Lugols iodine	Day of procedure	Enough to moisten the cervix
Cervical image capture	Day of procedure	3-5 images
Provider usability survey	End of study	6-8 minutes

7.3 Study Methodology /Study Procedures

Clinical data will be captured on the smart-phone by clinician performing papsmear or colposcopist.

This information will be captured on the study workflow.

Date	Study ID	Age	HIV status	Cytology (if known)	HPV status (if known)	Colposcopy impression	Histology result

Cervical images will be captured (at the time of planned papsmear, colposcopy/LLETZ/cone biopsy) using the smartphone system at three stages: pre-acetic acid, post-acetic acid, and post-Lugol's iodine. Histopathology and cytology results will be abstracted and linked using study IDs. Optional clinician impressions will be recorded.

Provider usability will be assessed using a survey.

All procedures will follow a standard operating protocol.

8. Study Population

8.1 Recruitment Procedure

Participants will be women aged 25–65 years referred for colposcopy-guided LLETZ, cone biopsy or presenting for a routine papsmear. They will be recruited during pre-procedure consultations. Recruitment will be led by the clinical team or trained research staff. Patients will be informed that participation is voluntary and will not affect care.

Recruitment at Mankweng Hospital will be led by the academic gynaecology team, while at Quality Care Private Hospital recruitment will occur within the outpatient service. The same eligibility criteria, consent process, and data-capture protocol will apply at both sites.

8.2 Inclusion Criteria

- Women aged 25-65 years undergoing colposcopy guided LLETZ procedure, cone biopsy or routine papsmear at participating sites
- Patients with a clinical indication for colposcopy as determined by an abnormal cervical screening test:
  1) Abnormal cytology
     i) Atypical Squamous Cells of Undetermined Significance (ASC-US)

ii) Atypical Squamous Cells, HSIL cannot be excluded (ASC-H)
iii) Low Grade Squamous Intraepithelial Lesion (LSIL)
iv) High Grade Squamous Intraepithelial Lesion (HSIL)

2) High risk HPV positive

- Sufficient visibility of the cervix during the examination to permit imaging

8.3 Exclusion Criteria

- Previous cervical surgery (e.g. LLETZ/cone biopsy)
- Pregnant women, due to altered cervical appearance and ethical considerations
- Presence of cervical polyps or cervical benign neoplasms
- Patients with active vaginal bleeding at the time of the procedure, which could obscure the cervix and affect image quality
- Inability or unwillingness of participant or legally acceptable representative to give written informed consent.

8.4 Consent

Consent Overview

Method of obtaining informed consent:

Individual informed consent will be obtained from each participant prior to any research procedures. Written consent will be obtained following a verbal explanation of the study and provision of the Participant Information Sheet and Consent Form (PICF).

Fundamental conditions for valid informed consent:

1) Disclosure of relevant information
All prospective participants will receive a detailed explanation of the study's purpose, procedures, potential risks and benefits, data handling and privacy safeguards. This information will be provided both verbally by the research team and in writing through the PICF, available in English and any local languages as required. Participants will be informed that their participation is voluntary and that they may withdraw any time without affecting their clinical care.

2) Comprehension of the information provided
The research team will ensure that participants understand the study by;

a) Allowing adequate time to read the PICF
b) Encouraging questions and clarifying any concerns
c) Using interpreters if needed to ensure full understanding
d) Assessing comprehension through open-ended questions

3) Voluntary agreement free from coercion

Participation will be entirely voluntary. Consent will be obtained in a non-pressured environment, separate from clinical treatment areas, and participants will be reminded that their decision will not affect their current or future medical care. No incentives or undue influence will be applied.

Provision of information and consent process:

- Consent will be sought during the pre-procedure consultation (on the same day as the papsmear, LLETZ or cone biopsy where possible).
- Participants will be informed that cervical images will be captured using a smartphone-based device and that these images, along with their anonymized histopathology and cytology results, will be used to train and validate an AI model.
- They will be informed that no personally identifying images or information will be stored
- They will be made aware of any data storage de-identification and access protocols

Participant commitment, project duration and follow-up:

- Participant commitment

There will be minimal participant commitment. Participation involves the capture of cervical images during a procedure (papsmear/LLETZ/cone biopsy) already scheduled as part of their clinical care. Additional procedures include application of acetic acid for those undergoing routine papsmear.

- Project duration

The study will run over 6-12 months

- Follow up

No follow-up is required from participants beyond their routine post-procedural care. All histology and cytology results will be sourced from the standard pathology reports.

Re-consent:

Re-consent will be required if:

- There are substantial protocol changes that affect the scope, risks, or use of the data.
- Future use of data/images is proposed beyond the scope of the original consent.

8.5 Risk Management and Safety

Risks are minimal and primarily related to psychological discomfort or data privacy. Image capture is non-invasive. Emotional support will be available. All data are de-identified and securely stored. Contingency plans include additional site recruitment and weekly image quality audits.

8.6 Handling of Withdrawals

Participants may withdraw at any time. Previously collected data may be retained unless the participant requests otherwise. Withdrawals will be documented.

8.7 Replacements

Withdrawn participants will not be replaced. The sample size includes a buffer to maintain statistical power.

9. Data analysis

9.1 Analysis of quantitative data

AI model performance will be evaluated using sensitivity, specificity, PPV, NPV, and AUROC. McNemar's test, Cohen's kappa, and DeLong's method will be used for comparisons. Sample size of 500 provides >90% power to detect a 10% difference at $\alpha = 0.05$

9.2 Analysis of qualitative data

Provider survey responses will be analysed using thematic analysis aligned with the Technology Acceptance Model. NVivo software will be used.

10. Data Management

10.1 Data Generation (source data)

Source data

Generated from patient files (demographic and clinical information), image capture, histopathology reports, cytology reports, and survey responses.

Source data from both Mankweng Hospital and Quality Care Private Hospital will follow identical documentation and coding procedures to ensure harmonised data capture across sites.

10.2 Data Capture Methods and Data Use, Storage, Access & Disclosure

Data collection

Meta data (patient age, HIV status, HPV/cytology results and colposcopy impression) will be captured on our workflow in CSV format.
Images will be captured using a specially designed clip-on lens attached to a smartphone and transferred in JPEG format to the workflow.
Data pre-processing will be conducted (resizing/cropping, colour normalization, noise reduction, image augmentation)
Meta data cleaning will be done to ensure structured formatting, handle missing values and normalize categorical variables
Data will be processed through annotation, feature engineering (combine image features with meta data) and data splitting (train 70%,validation 15%, test sets 15%).

Each site will capture data locally using the same smartphone-based device. Data will first be stored on password-protected local drives within each facility and uploaded weekly to the central encrypted cloud server hosted in compliance with POPIA. No identifiable information will leave either site. Site-specific folders will maintain audit trails to distinguish data origin while ensuring uniform data structure.

Data storage and access

Data will be encrypted in transit and at rest.
Stored on secure cloud servers compliant with POPIA Act.

Access to data

Access restricted to authorized personnel only (clinicians, researchers)
On device access will be password protected.

Cloud access will require username and strong password with two-factor authentication.
Access control will be role-based e.g.;
Community health worker-capture and view own patients
Clinical supervisor- review regional patients
Data scientist- anonymized data only
Administration- system configuration only.

No personally identifiable information or data will be released to third parties.

10.3 Data Confidentiality

Personal information and data will be de-identified, with the study ID linking the patient to the histopathology and cytology results. We do acknowledge the potential to re-identify participants through linking study ID to pathology results. Such risks will be minimized by restricting access to the final data-set to the researchers.

Once data capturing is complete, study ID's will be replaced with a different code for the AI engineer to train, validate and test the model. In this way, only the researchers will have access to the full data-set linking patients with pathology results through the study ID. After model training and testing, data will be stored without the study ID.

Study ID's will also be linked to informed consent forms. These forms will only be accessible to researchers.

Both Mankweng Hospital and Quality Care Private Hospital will designate an on-site data custodian responsible for ensuring adherence to confidentiality and secure transfer protocols.

10.4 Archiving – Data and Document Retention

Archiving

Retention periods

Data type	Retention period	Justification
De-identified cervical images	10 years	For training and validating AI systems and academic publication
Patient metadata (linked to image)	5 years	For statistical research, de-identified and access restricted
Signed consent forms	7 years	Regulatory and ethical audit trail
Audit logs/access records	2 years	Security monitoring and compliance review

All data will be encrypted at rest and in transit.
Smartphone devices are password protected.
Cloud storage complies with POPIA and will be located in approved jurisdictions.
Informed consent forms will be uploaded onto a patient database and disposed of at the end of the retention period. Hard copies will be disposed of via a confidential shredding process.
Researchers (Lucia Chimanga and Claris Tigere) will be the custodians during the archive period.
Records will not be disposed of without the written consent of the Coordinating Principal Investigator

11. Appendix

List of Attachments included:

Document Name	**Version Number**	**Date** (e.g. 18 January 2021)
Survey	1.0	11 June 2025
Consent Form	1.0	23 September 2025

11.1 Survey

Post-Study Clinician Survey: AI Colposcopy Usability and Acceptability

(This survey will take approximately 6-8 minutes to complete)

Purpose:
This survey aims to gather feedback from clinicians on their experience using the smartphone-based colposcopy device. Your input will help refine the tool, improve user experience, and guide implementation strategies.

Section A: Background Information

1. Your professional role (tick where appropriate):

☐ Consultant Gynaecologist

☐ Registrar / Trainee

☐ Nurse Colposcopist

☐ Other (please specify): _____

2. Years of colposcopy experience:

☐ <1 year

☐ 1–5 years

☐ >5 years

3. Type of facility where you used the device:

☐ Tertiary hospital

☐ Secondary hospital

☐ Primary healthcare facility

☐ Private

4. Approximate number of patients examined with the device during the study:

☐ 1–10

☐ 11–30

☐ >30

Section B: Usability of the Device

Scale: 1 = Strongly Disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly Agree

Statement	1	2	3	4	5
The device was easy to assemble and set up.	☐	☐	☐	☐	☐
The smartphone interface was intuitive and user-friendly.	☐	☐	☐	☐	☐
Image capture was straightforward and consistent.	☐	☐	☐	☐	☐
The lens provided clear visualization of the cervix.	☐	☐	☐	☐	☐
The lighting source was adequate for examination.	☐	☐	☐	☐	☐
The device was comfortable to use during the examination.	☐	☐	☐	☐	☐
Using the device did not significantly prolong examination time.	☐	☐	☐	☐	☐
I would feel confident using this device	☐	☐	☐	☐	☐

independently in clinic.

Section C: Workflow Integration

Scale: 1 = Strongly Disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly Agree

Statement	1	2	3	4	5
Using the device fit smoothly into my usual clinic workflow.	☐	☐	☐	☐	☐
It did not cause significant delays or disruptions in patient flow.	☐	☐	☐	☐	☐
Image uploading was quick and efficient.	☐	☐	☐	☐	☐
I would be happy to integrate this system into regular clinic practice.	☐	☐	☐	☐	☐

Section D: Perceived Impact on Patient Care

Scale: 1 = Strongly Disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly Agree

Statement	1	2	3	4	5
This tool has the potential to improve diagnostic accuracy if supported by AI diagnostic capabilities.	☐	☐	☐	☐	☐
It could help standardize colposcopy interpretation between clinicians.	☐	☐	☐	☐	☐
It could increase access to colposcopy in low-resource or rural settings.	☐	☐	☐	☐	☐
Patients appeared comfortable during examinations with this device.	☐	☐	☐	☐	☐
I believe an AI-assisted device can improve early detection of cervical pre-cancer.	☐	☐	☐	☐	☐

Section E: Data Privacy, Ethics and Responsibility

Scale: 1 = Strongly Disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly Agree

Statement	1	2	3	4	5
I am confident that patient data captured by the system are stored securely.	☐	☐	☐	☐	☐
I believe the system maintains patient confidentiality.	☐	☐	☐	☐	☐
I would like more transparency on how the AI model is trained and validated.	☐	☐	☐	☐	☐

Section H: Overall Satisfaction

Scale: 1 = Strongly Disagree | 2 = Disagree | 3 = Neutral | 4 = Agree | 5 = Strongly Agree

Statement	1	2	3	4	5
Overall, I am satisfied with my experience using the device.	☐	☐	☐	☐	☐
I would recommend this system	☐	☐	☐	☐	☐

to colleagues.					
I would like to continue using this system beyond the study period.	☐	☐	☐	☐	☐

On a scale of 0–10, how likely are you to recommend this device and AI system to a colleague?

☐0 ☐1 ☐2 ☐3 ☐4 ☐5 ☐6 ☐7 ☐8 ☐9 ☐10

Section I: Open-Ended Feedback

1. What aspects of the device or AI system did you find most useful?

_____

_____

_____

2. What challenges or limitations did you experience?

_____

_____

_____

3. What improvements would you suggest for future versions?

_____

_____

_____

4. Any additional comments or feedback:

_____

_____

_____

11.2 Consent Form

English Version

PATIENT CONSENT FORM

Study: Using Pictures to Improve Cervical Cancer Diagnosis with AI

Research Centre: University of Limpopo/Mankweng Hospital and Quality Care Private Hospital

Researcher: Dr. Lucia Chimanga, Dr. Claris Tigere

Why are we doing this study?

Cervical cancer is a big health problem for women in South Africa. We are testing a computer program called ColpoAI that can help doctors find problems earlier and more accurately.

To do this, we need to take pictures of the cervix (the opening to the womb) during normal procedures you are already having (papsmear, colposcopy or LLETZ).

What will happen?

During your procedure, the doctor will take pictures of your cervix. These pictures will be stored safely and used only for research. No extra tests or risks will be added to your care. If you are having a papsmear, we may need to apply acetic acid to your cervix so we can see the cells better. Acetic acid will not cause any damage to your cervix or body.

What are the benefits of this?

There may be no direct benefit to you. Your participation will help improve cervical cancer care for other women in South Africa in the future.

Risks

No extra medical risks beyond your normal procedure. We will protect your privacy. Your name will not be linked to your pictures.

Your choice

Taking part is your choice.

You may say no or stop at any time.

Your treatment at the practice will not be affected in any way.

You will not receive money for participating.

Consent

I understand what this study is about. I agree that pictures of my cervix may be taken during my procedure and used for research to improve cervical cancer care in South Africa.

Patient Name: _____

Signature/Thumbprint: _____ Date: _____

Witness/Researcher Name: _____

Signature: _____ Date: _____

Tshivenda Version

Study: Using Pictures to Improve Cervical Cancer Diagnosis with AI

Research Centre: University of Limpopo/Mankweng Hospital and Quality Care Private Hospital

Researcher: Dr. Lucia Chimanga, Dr. Claris Tigere

Purpose of the Study / Tshikalo tsha Nyambo

English: Cervical cancer is a major health problem in South Africa. We are testing ColpoAI, a computer program that helps doctors detect cervical problems earlier and more accurately. During your routine colposcopy or LLETZ, we would like to take pictures of your cervix for research.

Tshivenda: Kankere ya tshikovha ndi thaidzo khulwane kha Afrika Tshipembe. Ri khou lingedza khomphutha ya ColpoAI ine ya thusa dokotela u wana thaidzo nga u ṱavhanya na nga vhuimo ho teaho. Nga tshifhinga tsha colposcopy kana LLETZ, ri humbela u dzhia dzifhoto dza tshikovha tsha inwi kha nyambo.

What Will Happen / Zwi do Itwa mini

English: Pictures of your cervix will be taken during your procedure. These will be stored securely and used only for research. No extra tests or risks will be added.

Tshivenda: Dzifhoto dza tshikovha tsha vho dzi do dzhiiwa nga tshifhinga tsha vhuṱanzi. Dzoṱhe dzi do vhulungwa nga vhuṱali na u shumiswa fhedzi kha nyambo. A huna zwiitea zwinzhi kana khombo.

Additional Note / Tshivhumbeo tshiswa:

English**:** Patients undergoing Pap smears will have acetic acid applied to the cervix as an additional step, which would not normally be done if they were not part of this study.

Tshivenda: Vhafumakadzi vha no ita Pap smear vha do dzheniselwa na u swikisiwa acetic acid kha tshikovha tshavho sa tshiteṅwa tshiswa, tshe a tshi nga itwi arali vha si tshifhinga tsha nyambo.

Benefits / Muhola

English: You may not benefit directly, but your participation will help improve cervical cancer diagnosis for women in South Africa.

Tshivenda: A hu na muhola muhulwane kha inwi, fhedzi u dzhenela hau hu do thusa u khwinisa u wanululwa ha kankere ya tshikovha kha vhafumakadzi vha Afrika Tshipembe.

Risks / Khombo

English: No extra risks beyond your usual procedure. Your name will not be linked to your images.

Tshivenda: A huna khombo dza engetelekaho kha tshifhinga tsha u tshimbilela. Dzina ḽa vho a ḽi nga tumani na zwifanyiso dza vho.

Your Choice / Khetho yau

English: Participation is voluntary. You may refuse or stop at any time. Your hospital care will not be affected. No money will be given.

Tshivenda: U dzhenela ndi nga vhulofhedzi. Vha nga hana kana vha litsha nga tshifhinga tshinwe na tshinwe. Dzilafho dza vho dzi do vha vhukati ha vho na tshibadela. A huna tshelede ine ido waniwa.

Consent / Mulandu wa u Tenda

English: I understand the study and agree to allow pictures of my cervix to be taken during my procedure for research.

Tshivenda: Ndo pfesesa hoyu nyambo nahone ndi khou tenda uri dzifhoto dza tshikovha tshanga dzi dzhiiwe nga tshifhinga tsha vhuṱanzi uri dzi shumiswe kha nyambo.

Patient Name / Dzina ḽa Murwayi: _____

Signature/Thumbprint / Saini kana Munwe: _____ Date / Duvha: _____

Researcher/Witness / Muvhumbudzi: _____

Signature / Saini: _____ Date / Duvha: _____

Xitsonga Version

Study: Using Pictures to Improve Cervical Cancer Diagnosis with AI

Research Centre: University of Limpopo/Mankweng Hospital and Quality Care Private Hospital

Researcher: Dr. Lucia Chimanga, Dr. Claris Tigere

Purpose of the Study / Njhekanjhekisiso wa Ndzavisiso

English: Cervical cancer is a major health problem in South Africa. We are testing ColpoAI, a computer program that helps doctors detect cervical problems earlier and more accurately. During your routine colposcopy or LLETZ, we would like to take pictures of your cervix for research.

Xitsonga: Kankere ya xivelekelo i xiphiqo lexikulu eAfrika Dzonga. Hi le ku kambeleni hi ColpoAI, porogiramu ya khompyuta leyi nga pfunaka madokotela ku kuma swivangelo hi nkarhi na hi ndlela yo kongoma. Hi nkarhi wa colposcopy kumbe LLETZ, hi ta teka swifaniso swa xivelekelo ra wena leswaku swi tirhisiwa eka ndzavisiso.

What Will Happen / Leswi Swi Nga Ta Endleka

English: Pictures of your cervix will be taken during your procedure. These will be stored securely and used only for research. No extra tests or risks will be added.

Xitsonga: Swifaniso swa xivelekelo na wena swi ta tekiwa hi nkarhi wa vukamberi. Swi ta hlayisiwa hi vuhlayiseki naswona swi ta tirhisiwa ntsena eka ndzavisiso. A ku na vukamberi byin'wana kumbe khombo.

Additional Note / Nkombiso Lowuntshwa:
English**:** Patients undergoing Pap smears will have acetic acid applied to the cervix as an additional step, which would not normally be done if they were not part of this study.
Xitsonga: Varhambi lava endlaka Pap smear va ta n'wi rhumela asidi ya acetic ehenhla ka

khwiri ra vona tanihi vuhlayiseki byo engeteleka, lebyi a byi nga endliwi loko va nga ri xiphemu xa ndzavisiso lowu.

Benefits / Vantlhanu

English: You may not benefit directly, but your participation will help improve cervical cancer diagnosis for women in South Africa.

Xitsonga: U nga ha nga vi na muholo wo kongoma, kambe ku nghena ka wena ku ta pfuna leswaku ku antswisiwa ku tsarisa kankere ya khwiri eka vavasati va Afrika Dzonga.

Risks / Swiphiqo kumbe Khombo

English: No extra risks beyond your usual procedure. Your name will not be linked to your images.

Xitsonga: A ku na khombo rin'wana handle ka vuvhambi bya wena bya ntolovelo. Vito ra wena a ri nge hlanganisiwi ni swifaniso swa wena.

Your Choice / Xikombiso xa wena

English: Participation is voluntary. You may refuse or stop at any time. Your hospital care will not be affected. No money will be given.

Xitsonga: Ku nghena ka wena i nga ku rhandza ka wena. U nga ala kumbe u tshika hi nkarhi wihi na wihi. Vuvhambi bya wena e xibedhlele byi ta tshama byi fanile. A wu nge amukeli mali.

Consent / Ku Tisa Mbilu

English: I understand the study and agree to allow pictures of my cervix to be taken during my procedure for research.

Xitsonga: Ndzi twisisa ndzavisiso lowu naswona ndzi pfumela leswaku swifaniso swa xivelekelo xa mina swi tekiwa hi nkarhi wa vukamberi leswaku swi tirhisiwa eka ndzavisiso.

Patient Name / Vito ra Murhambi: _____

Signature/Thumbprint / Saini kumbe Munwe: _____ Date / Siku: _____

Researcher/Witness / Mulavisisi: _____

Signature / Saini: _____ Date / Siku: _____

Sepedi Version

PATIENT CONSENT FORM

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Sepedi Translation

Ke ka lebaka la eng re dira nyakišišo ye?

Mokgwapetšo wa popelo ke bothata bjo bogolo bja maphelo a basadi mo Afrika Borwa. Re leka lenaneo la khomphutha le le bitšwago ColpoAI le le ka thušago dingaka go hwetša mathata kapele le ka nepo.

Go dira se, re hloka go tšea diswantšho tša mokgwapetšo wa popelo nakong ya kalafo yeo o šetšego o e dirwa (colposcopy goba LLETZ).

What will happen?

During your procedure, the doctor will take pictures of your cervix.

These pictures will be stored safely and used only for research.

No extra tests or risks will be added to your care.

In addition, patients undergoing pap smears will have acetic acid applied to the cervix as an additional procedure, which would not be added if they were not participating in the study.

Sepedi:

Nakong ya kalafo, ngaka e tla tšea diswantšho tša molomo wa popelo ya gago.

Diswantšho tše di tla bolokwa ka tshireletšo gomme di šomišwa fela bakeng sa nyakišišo.

Ga go na diteko goba kotsi tše dingwe tše mpsha tšeo di tla oketšwago tlhokomelong ya gago.

Gape, balwetsi bao ba dirwago pap smear ba tla tsenelwa ka acetic acid popelong bjalo ka tiragalo ye nngwe e kgethegilego, yeo e ka bego e se ya dirwa ge nkabe ba sa tšea karolo mo nyakišišong.

Will this help me?

There may be no direct benefit to you.

Your participation will help improve cervical cancer care for other women in South Africa in the future.

Sepedi:

Ga gona tefo yeo o ka e lwetsago.

Go tšea karolo ga gago go tla thuša go kaonafatša tlhokomelo ya mokgwapetšo wa popelo bakeng sa basadi ba bangwe mo Afrika Borwa nakong e tlago.

Risks

No extra medical risks beyond your normal procedure.

We will protect your privacy. Your name will not be linked to your pictures.

Sepedi:

Ga go na kotsi ye nngwe ya kalafo ka ntle ga kalafo ya gago ya tlwaelo. Re tla šireletša sephiri sa gago. Leina la gago le ka se tswalanywe le diswantšho tša gago.

Your choice

Taking part is your choice.

You may say no or stop at any time.

Your treatment at the hospital will not be affected in any way.

You will not receive money for participating.

Sepedi:

Go tšea karolo ke kgetho ya gago.

O ka bolela aowa goba o emiše nako efe goba efe.

Kalafo ya gago sepetlele e ka se amege ka tsela efe goba efe.

O ka se hwetše tšhelete ka lebaka la go tšea karolo.

Consent

I understand what this study is about. I agree that pictures of my cervix may be taken during my procedure and used for research to improve cervical cancer care in South Africa.

Sepedi:

Ke kwešiša gore nyakišišo ye e mabapi le eng. Ke dumelelana gore diswantšho tša mokgwapetšo wa ka di tšeege nakong ya kalafo ya ka gomme di šomišwe bakeng sa nyakišišo go kaonafatša tlhokomelo ya mokgwapetšo wa popelo mo Afrika Borwa.

Patient Name / Leina la Mokuli: _____

Signature/Thumbprint / Saini goba Monwana: _____

Date / Letšatši: _____

Witness/Researcher Name / Leina la Mohlatswi goba Moithuti:
_____

Signature / Saini: _____ Date / Letšatši: _____

12. References

1. Bruni L, Albero G, Amarilla S et al. Human Papillomavirus and Related Diseases in South Africa. Summary Report March 2023.2021: Available from https://hpvcentre.net/statistics/reports/ZAF

2. Bray F, Ferlay J, Soerjomataram I, Siegel RL et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: A Cancer Journal for Clinicians.2018:68(6):394-424

3. IARC Working Group on the Evaluation of Cancer-Preventive Strategies. Cervix Cancer Screening. IARC Handbooks of Cancer Prevention, Volume 10. Lyon, France: International Agency for Research on Cancer; 2005.

4. Lei J, Plomer A, Elfstrom M et al. HPV Vaccination and the Risk of Invasive Cervical Cancer. N Engl J Med. 2020 Sep;383(14):1340-1348.

5. World Health Organization. Global strategy to accelerate the elimination of cervical cancer as a public health problem. Geneva: WHO; 2020. Available from: https://www.who.int/initiatives/cervical-cancer-elimination-initiative

6. World Health Organization. WHO guideline for screening and treatment of cervical pre-cancer lesions for cervical cancer prevention. Geneva: WHO; 2021. Available from: https://www.who.int/publications/i/item/978920030824

7. World Health Organization. WHO guideline for screening and treatment of cervical pre-cancer lesions for cervical cancer prevention: Use of dual-stain cytology to triage women after a positive test for human papillomavirus (HPV). Geneva: WHO; 2024. Available from: https://iris.who.int/bitstream/handle/10665/376492/9789240091658-eng.pdf

8. Botha M, Mabenge M, Makua M et al. Cervical cancer screening guidelines for South Africa. Afr J Obstet Gynaecol. 2023;1(1).

9. National Department of Health. Cervical cancer prevention and control policy. Pretoria: National Department of Health; 2017.

10. Xue P, Ng M, Qiao Y et al. The challenges of colposcopy for cervical cancer screening in LMICs and solutions by artificial intelligence. BMC Med. 2020;18:169. doi:10.1186/s12916-020-01613-x

11. Tan J, Neesham D, Szabo R et al. Colposcopy performance in the new primary HPV screening in Australia: How to determine colposcopy competency? Aust N Z J Obstet Gynaecol. 2024;64:375-82.

12. Domgue J, Dille I, Gnangran F et al. Utility of colposcopy for the screening and management of cervical cancer in Africa: a cross-sectional analysis of provider's training and practices. BMC Health Serv Res. 2024;24:1619. doi:10.1186/s12913-024-11982-1

13. Katz IT, Butler L, Crankshaw T et al. Cervical abnormalities in South African women living with HIV with high screening and referral rates. J Glob Oncol. 2016;2:375-80. doi:10.1200/JGO.2015.002469

14. Maimela G, Nene X, Mvundla N, Sawry S, Smith T, Rees H, et al. The impact of decentralising colposcopy services from tertiary level to primary level care in inner city Johannesburg, South Africa: a before and after study. BMJ Open. 2019;9(3):e024726. doi:10.1136/bmjopen-2018-024726

15. Alowais S, Alghamdi S, Alsuhebany N et al. Revolutionizing healthcare: the role of artificial intelligence in clinical practice. BMC Med Educ. 2023;23:689. doi:10.1186/s12909-023-04698-2

16. Madathil S, Dhazib M, Lelong Q et al. A multimodal deep learning model for cervical pre-cancer and cancers prediction: Development and internal validation study. Comput Biol Med. 2025;186:109710.

17. Aquilina A, Papagiannakis E. Deep learning diagnostic classification of cervical images to augment colposcopic impression. J Low Genit Tract Dis. 2024;28(3).

18. Chang C, Xu Y, Wang X, et al. Deep learning-based classification of cervical images for cancer screening: A comparative study. Comput Biol Med. 2022;142:105205.

19. De Fauw J, Ledsam JR, Romera-Paredes B et al. Clinically applicable deep learning for diagnosis and referral in retinal disease and beyond. Nat Med. 2023;29(1):9–17.

20. Mascarenhas M, Alencoao I, Carinhas M et al. Artificial intelligence and colposcopy: automatic identification of cervical squamous cell carcinoma precursors. J Clin Med. 2024;13:3003. doi:10.3390/jcm3103003

21. Talathi MA, Dabhadkar S, Doke P, et al. Accuracy of the AI-Based Smart Scope Test as a point-of-care screening and triage tool compared to colposcopy: A pilot study. Cureus. 2025;17(3):e81212. doi:10.7759/cureus.81212

22. Qin ZZ, Van der Walt M, Moyo S, Ismail F, Maribe P, Denkinger CM, et al. Computer-aided detection of tuberculosis from chest radiographs in a tuberculosis prevalence survey in South Africa: external validation and modelled impacts of commercially available artificial intelligence software. Lancet Digit Health. 2024;6(6):e400–11. doi:10.1016/S2589-7500(24)00118-3